

◆ NETWORK MONITOR — COMPLETE U

Network Monitor

Real-time network intelligence — speed
visualisation, and AI-powered analysis in

What is Network Monitor?

Network Monitor is an all-in-one Windows desktop application for real-time network analysis, built around a dark futuristic interface with live gauges, packet capture, topology mapping, and AI-assisted investigation.



Speed Test Dashboard

Live animated gauges showing download, upload, and ping with 5-minute automatic testing and historical charts.



Wireshark Monitor

Full packet capture frontend with display filters, hex dump, stream following, statistics, and AI query.



EtherApe Topology

Real-time network topology visualiser showing hosts and flows with live traffic animation.



DNS Monitoring

Continuous DNS resolution timing across 6 major hosts with history chart and jitter tracking.



AI Analysis

Ask Claude to explain captures in plain English. Generate professional HTML reports with multiple queries.



Data Export

Export speed test history as CSV or JSON with flexible date range selection and folder picker.

Installation

The installer handles all dependencies automatically. An internet connection is required during installation to download Wireshark and Nmap.

1

Run NetworkMonitorSetup.exe as Administrator

Right-click the installer and choose "Run as administrator". This is required to install Wireshark's Npcap driver.

2

Choose components

All components are selected by default. Wireshark + Npcap is required for packet capture. Nmap enables network scanning features.

3

Wait for downloads

Wireshark (~80 MB) and Nmap (~30 MB) will be downloaded and installed silently. This takes 2–5 minutes depending on connection speed.

4

Launch from Desktop or Start Menu

A shortcut is created on your Desktop and in Start Menu → Network Monitor.



Administrator rights required for capture. Packet capture (Wireshark Monitor and EtherApe) requires the application to be run as Administrator. Right-click the shortcut → Run as administrator.

System Requirements

COMPONENT	MINIMUM
OS	Windows 10 64-bit or later
RAM	4 GB (8 GB recommended for heavy captures)
Disk	500 MB free (plus ~200 MB for Wireshark)
Network	Any wired or wireless adapter supported by Npcap
Internet	Required for speed tests and AI analysis

First Run

On first launch the application runs an initial speed test automatically. The main window displays five animated gauges across the top and six charts below.

i Speed test data is saved to a JSON file in your system temp folder and persists between sessions. You will see historical charts fill in over time as more tests run.

The app auto-runs a speed test every **5 minutes** in a background thread. You can trigger one manually at any time with the  button.

Dashboard Layout

The main window is divided into three horizontal bands: the gauge row at the top, a button bar in the middle, and six charts at the bottom.

Window Regions

REGION	CONTENTS
Gauge row (top)	Five circular dials — Download, Upload, Ping, DNS, and the View Mode selector. All animate smoothly with easing.
Button bar (middle)	Export Data, Run Speed Test, Monitor DNS, Wireshark, EtherApe buttons. Colour swatches on the far right.
Chart area (bottom)	Six charts in a 2×3 grid: Download, Upload, Latency, Daily Averages, DNS History, and Statistics table.

Speed Gauges

The five dials at the top of the window provide an at-a-glance view of your current network performance. Each is a 270° neon sweep gauge with animated needles.

GAUGE	UNIT	WHAT IT SHOWS
Download ↓	Mbps	Last measured download bandwidth. Scale auto-adjusts to 130% of your historical maximum.
Upload ↑	Mbps	Last measured upload bandwidth.
Ping ○	ms	Round-trip latency from the most recent speed test.
DNS ○	ms	Average DNS resolution time across 6 hosts from the last DNS check. Updated only when you click Monitor DNS.
View Mode	—	Rotary selector for chart time window. Click a segment to change view.



The needles ease smoothly toward their target values. The large digital readout in the centre of each dial shows the exact current value.

Needle Easing

Gauges update every second with a 20% easing factor — the needle moves 20% of the remaining distance each tick, giving a smooth analog feel. On a manual "Run Speed Test", the needle snaps immediately to the new value.

History Charts

CHART	DESCRIPTION
Download	Bar chart (Today/History view) or neon line chart (weekly/monthly). Shows all tests in the current time window.
Upload	Same layout as Download, using the Upload colour.
Latency	Ping values for the selected time window.
Daily Averages	Always shows the last 14 days regardless of view mode. Grouped bars for Download and Upload side by side.
DNS History	Time-series chart of all DNS check results. Shows 20 ms and 100 ms reference lines for quick assessment.
Statistics	Table of Avg / Max / Min / σ / Jitter for Download, Upload, Ping, and DNS in the current view window.

i **Jitter** is calculated as the mean absolute difference between consecutive ping samples — the standard VoIP/RFC 3550 definition. High jitter causes choppy audio/video calls.

View Modes

Click any segment of the View Mode dial (top-right) to change the time window displayed in the three main charts and the Statistics table.

MODE	TIME WINDOW	CHART STYLE
Today	Midnight to now	Hourly bar chart (0:00–24:00 x-axis)
This Week	Start of current week to now	Neon line chart with date labels
This Month	1st of current month to now	Neon line chart with date labels
All Time	All recorded data	Neon line chart

Running Speed Tests

Speed tests run automatically every 5 minutes. To run one immediately, click

 in the button bar.

The button label changes to  during the test. Results appear in the gauges and charts as soon as the test completes (typically 20–60 seconds).

How Speed Tests Work

The app uses the **Ookla Speedtest CLI** () with three fallback attempts:

- 1 JSON mode**
Fastest and most accurate. Returns structured data directly.
- 2 Plain text mode**
Falls back to parsing human-readable output if JSON fails.
- 3 Bare run**
Last resort — runs speedtest.exe with no flags and parses any output found.

 Speed test results require an active internet connection. If all three attempts fail, check that  is present in the install folder and that your firewall is not blocking it.

DNS Monitoring

Click  to run a DNS resolution check. The app resolves these six hosts and measures the wall-clock time for each:

- google.com
- cloudflare.com
- microsoft.com
- amazon.com
- bbc.co.uk
- github.com

The average of all successful resolutions is recorded and displayed on the DNS dial and in the DNS History chart. Up to 500 readings are retained.

Interpreting DNS Times

RANGE	ASSESSMENT
< 20 ms	Excellent — fast local or ISP DNS
20–50 ms	Good — normal for most connections
50–100 ms	Moderate — consider using 8.8.8.8 or 1.1.1.1
> 100 ms	Slow — DNS may be causing browsing delays



High DNS times can make websites feel slow even when your broadband speed is fine. Try setting your DNS server to `1.1.1.1` (Cloudflare) or `8.8.8.8` (Google) in your network adapter settings.

Exporting Data

Click ↓ EXPORT DATA to open the export dialog.

Export Dialog Options

OPTION	DESCRIPTION
From / To date	Set the date range manually (YYYY-MM-DD HH:MM:SS format)
Quick range	Today, 7 days, 30 days, or All — fills date fields instantly
Format	CSV (Excel-compatible) or JSON
Save to	Destination folder — defaults to app data folder, click Browse to change

Export File Contents

Each row / object in the export contains: timestamp, download_mbps, upload_mbps, ping_ms.

Themes & Colours

Three colour swatch buttons () appear on the far right of the gauge area, labelled CLR. Click any swatch to open a colour picker for that channel:

SWATCH	CONTROLS
↓ Download	Colour of download gauge, line, and bars
↑ Upload	Colour of upload gauge, line, and bars
○ Ping	Colour of ping gauge and latency chart

Custom colours are saved to `speedtest_config.json` and persist between sessions.

Built-in Themes

- Ocean** — Teal / Purple / Amber (default)
- Sunset** — Coral / Yellow / Green
- Neon** — Neon green / Hot red / Cyan
- Pastel** — Soft blue / Lavender / Peach
- Mono** — Grey scale

Clicking any CLR swatch overrides the current theme with a custom colour and stores it as a custom palette. To return to a built-in theme, the config file must be edited or deleted.

Wireshark Monitor

A full-featured dark-themed packet capture and inspection tool built on top of tshark (Wireshark's command-line engine). Accessed via the  **WIRESHARK** button on the main dashboard.



Live Capture

Select any network interface and start capturing packets in real time with colour-coded protocol rows.



Display Filters

Filter the visible packet list using Wireshark filter syntax without stopping the capture.



Packet Detail

Click any packet to see the full decoded protocol tree and hex dump side by side.



Statistics

Protocol hierarchy, conversations, endpoints, I/O graph, DNS stats, HTTP stats, and more.



PCAP Files

Open existing PCAP/PCAPNG files or save the current capture for later analysis.



AI Query

Ask Claude to explain what happened in the capture in plain English and export a PDF report.

Starting a Capture

1

Select an interface

Choose your network adapter from the Interface dropdown. Wi-Fi and Ethernet adapters are listed with friendly names. Run as Administrator if the list is empty.

2

(Optional) Set a Capture Filter

Enter a BPF filter in the Capture Filter field before starting. Examples: `port 80`, `host 192.168.1.1`, `tcp`. This filters at the driver level and reduces overhead.

3

Click ► Start Capture

Packets appear in the list in real time, colour-coded by protocol. The packet counter updates in the toolbar.

4

Click ■ Stop when done

The capture is saved automatically as a PCAP file. The status bar shows how many packets were recorded.



Administrator required. Packet capture will fail silently or show an "access denied" error without admin rights. Right-click the application shortcut and choose "Run as administrator".

Toolbar Buttons

BUTTON	ACTION
► Start Capture	Begin capturing on the selected interface
■ Stop	Stop the capture and save to PCAP
⦶ Clear	Clear all packets from the list and delete the PCAP file
📁 Open PCAP	Load an existing PCAP or PCAPNG file
💾 Save PCAP	Save the current capture to a chosen location
♦ AI Query	Open the AI analysis window
Auto-scroll	Toggle automatic scrolling to the latest packet

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Display Filters

Display filters narrow the visible packet list *without* stopping the capture or deleting data. They use standard Wireshark filter syntax.

Common Filter Examples

FILTER	SHOWS
<code>tcp</code>	All TCP packets
<code>udp</code>	All UDP packets
<code>dns</code>	All DNS queries and responses
<code>http</code>	Unencrypted HTTP traffic
<code>tls</code>	Encrypted TLS/HTTPS traffic
<code>ip.addr == 192.168.1.1</code>	Traffic to or from a specific IP
<code>ip.src == 10.0.0.5</code>	Traffic from a specific source IP
<code>tcp.port == 443</code>	Traffic on port 443 (HTTPS)
<code>icmp</code>	Ping packets
<code>frame.len > 1000</code>	Large packets only

Press or click to activate the filter. Click to remove it and show all packets.

Right-Click Quick Filters

Right-click any packet row to access quick filter options:

- ▶ Filter by Source IP — shows only traffic from that address
- ▶ Filter by Destination IP — shows only traffic to that address
- ▶ Filter by Protocol — shows only that protocol type
- ▶ Follow TCP/UDP Stream — reconstructs the full conversation

- ▶ Copy Source IP / Dest IP / Info — copies to clipboard

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Inspecting Packets

Click any row in the packet list to inspect it. The window splits into:

PANEL	CONTENTS
Packet Details (left)	Decoded protocol tree — Ethernet frame → IP header → TCP/UDP → application layer. Expand nodes to see individual fields.
Hex Dump (right)	Raw bytes of the packet in hex with ASCII representation. Offsets shown on the left. Null bytes are dimmed.

Column Sorting

Click any column header to sort the packet list by that field. Click again to reverse the sort. Sortable columns: No., Time, Source, Destination, Protocol, Length, Info.

Colour Coding

- HTTP / HTTP2 — neon green
- TLS / SSL — soft green
- DNS / mDNS — amber
- DHCP — yellow
- ICMP — coral red
- TCP — teal
- UDP — purple
- ARP — light blue
- Other — ice white

Statistics Menu

The Statistics menu provides detailed analysis of the current capture. All statistics windows are non-blocking — the capture continues while they are open.

STATISTICS WINDOW	DESCRIPTION
Summary	File size, packet count, duration, average rate and packet size
Protocol Hierarchy	Tree breakdown of all protocols with frame/byte counts
Conversations — IPv4/TCP/UDP	All conversations with bytes A→B, B→A, total, and duration
Endpoints — IPv4/TCP/UDP	All endpoints with total Tx/Rx packet and byte counts
I/O Graph	Throughput (bytes/s) and packet rate (pkt/s) over time, plotted with neon lines
Packet Lengths	Distribution of packet sizes across standard buckets
DNS Statistics	Query types, response codes, and timing
HTTP Statistics	Request methods, response codes, and hosts

Analyse Menu

FEATURE	DESCRIPTION
Expert Information	Automatic detection of errors, warnings, notes, and chat events — colour coded by severity
Follow TCP Stream	Reconstructs a TCP conversation in ASCII, showing client (teal) vs server (purple) data
Follow UDP/TLS Stream	Same as TCP Follow for UDP and TLS sessions
Decode As	Re-dissect a port as a different protocol (HTTP, TLS, DNS, etc.)

PCAP Files

Opening a PCAP

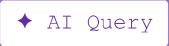
Click  and select any `.pcap`, `.pcapng`, or `.cap` file. The file is loaded and all packets are displayed. You can then apply display filters and run statistics on the file.

Saving a Capture

Click  to save the current capture to a chosen location. The file is always saved in PCAP format and can be opened in Wireshark, tcpdump, or any other compatible tool.

i During a live capture, tshark automatically writes packets to a temporary PCAP file in your system temp folder. This is what gets loaded when you click Save PCAP.

AI Query

Click  to open the AI analysis window. This sends a summary of your capture to Claude and lets you ask questions in plain English.

1

Enter your API key

Paste your Anthropic API key in the key field. Get one free at console.anthropic.com. Click  to remember it for future sessions.

2

Click a quick prompt or type your own

Six quick prompts are provided: Summarise, Suspicious patterns, Top hosts, DNS lookups, Network errors, Connections established.

3

Click Ask AI or press Enter

The capture summary (protocol breakdown, top IPs, sample packets, tshark stats) is sent to Claude. The response appears in the text area below.

What Context Is Sent

- Total packet count and capture duration
- Full protocol breakdown with counts
- Top 10 source and destination IPs by packet count
- Average packet rate
- First 40 and last 20 packets (full detail)
- tshark protocol hierarchy, IPv4 conversations, DNS tree



You do not need to stop the capture before using AI Query. The context is built from the packets currently visible in the list (including any active display filter).

Generating Reports

Build a multi-section HTML report from multiple AI queries:

- 1

Ask a question and get an AI response

Any question — summarise, security analysis, specific host investigation, etc.
- 2

Click + Add to Report

The question and answer are saved to the report queue. The counter updates: "Report: 2 entries".
- 3

Ask more questions and add them

Each query becomes its own section. Use the queue panel at the bottom to reorder (▲ ▼) or remove (X) entries.
- 4

Click  Generate Report

Choose a save location. An HTML file is created and opened in your browser automatically.
- 5

Print to PDF from your browser

In Chrome or Edge: File → Print → Destination: "Save as PDF". The report is print-optimised.

i Reports are self-contained HTML files — no internet connection needed to open them. They include the capture stats table, timestamps for each query, and all AI responses with markdown formatting rendered correctly.

EtherApe Monitor

A real-time network topology visualiser that shows hosts as nodes and network flows as animated lines. Accessed via the  button on the main dashboard.



Live Topology

Hosts appear as nodes sized by traffic volume. Flows are animated with pulse dots showing data direction.



Flow Inspection

Click any flow line on the canvas or in the table to see full IPs, protocol, packet count, data transferred, and average packet size.



PCAP Replay

Load a PCAP file and replay it at 0.1× to 16× speed (or unlimited) to visualise historical captures.



Font Control

Adjust font size (6–24) and family for all labels from toolbar 2. Changes apply to the whole window.

Live Capture

- 1

Select an interface

Choose your network adapter from the Interface dropdown. If the list is empty, the app needs Administrator rights.
- 2

(Optional) Set a BPF capture filter

Enter a filter like `tcp` or `port 80` in the Capture Filter field to limit traffic. Applied at capture start.
- 3

Click ► Start

The topology canvas populates as traffic is detected. Nodes and flows appear within seconds on an active network.
- 4

Click ■ Stop to end the capture

Existing nodes and flows fade gradually via traffic decay. Click  Clear to reset the view entirely.

Protocol Filter

The Protocol dropdown in toolbar 1 filters the *displayed* flows without stopping the capture. Options: ALL, TCP, UDP, DNS, HTTP, TLS, ICMP, ARP, DHCP.

Display Options

CHECKBOX	EFFECT
Labels	Show/hide IP address labels beneath each node
Legend	Show/hide the protocol colour legend in the top-left
Resolve DNS	Attempt to resolve IPs to hostnames (non-blocking, may take a moment)

Topology Canvas

Reading the Display

ELEMENT	MEANING
Node size	Proportional to cumulative bytes transferred (log scale). Larger = more traffic.
Node colour	Protocol of the most recent flow involving that host.
Line thickness	Proportional to flow data volume (log scale).
Pulse dots	Animated dots travelling along flow lines showing data direction.
Glow effect	Each flow line has a wide transparent glow for visibility against the dark background.

Traffic Decay

Flows and nodes that haven't received new packets gradually fade ($0.92\times$ decay per render tick at ~5 fps). This means inactive flows disappear from the canvas naturally, keeping the view focused on current activity.

Limits

LIMIT	VALUE
Maximum nodes	60 (oldest evicted when exceeded)
Maximum flows	150 (smallest flow evicted when exceeded)
Render rate	5 fps (200ms per frame — efficient, not heavy)
Table update rate	Every 2 seconds

Flow Details

Two ways to inspect a flow:

1 – Click a Flow Line on the Canvas

Click anywhere near a flow line in the topology view. The app calculates point-to-segment distance and selects the nearest flow within a 0.04 unit threshold. The detail panel populates automatically, and the matching row in the Active Flows table is highlighted.

2 – Click a Row in Active Flows Table

Click any row in the Active Flows table (bottom-right panel). The flow detail panel below the table updates with full information.

Flow Detail Panel Contents

FIELD	DESCRIPTION
Protocol	Protocol tag (colour-coded to match the flow)
Source IP	Full untruncated source IP address
→ hostname	Resolved hostname (if DNS Resolve is enabled)
Destination IP	Full untruncated destination IP address
Packets	Total packet count for this flow
Data (approx)	Approximate bytes transferred (B / KB / MB / GB)
Avg pkt size	Average bytes per packet

Resizing Panels

All three panel dividers (horizontal canvas vs tables split, hosts vs flows split, flows vs detail split) have raised visible sash handles. Drag them to resize. The panels remember their positions while the window is open.

PCAP Replay

Replay a previously captured PCAP file through the topology visualiser to investigate historical traffic.

1 **Click Open PCAP**

Browse for a .pcap, .pcapng, or .cap file. The filename appears in amber text in the toolbar once loaded.

2 **Set replay speed**

Use the Speed spinner: 0.1×, 0.25×, 0.5×, 1× (real-time), 2×, 4×, 8×, 16×, or 0 (unlimited speed — no timing delays).

3 **Click ▶ Replay**

The canvas clears and traffic replays. Progress shows in the status bar as "packet N / total (%)". The topology builds as packets are processed.

4 **Click ■ Stop Replay to halt**

The topology freezes at the point where replay was stopped. You can then inspect flows and run AI queries against the replayed data.

i Live capture and PCAP replay are mutually exclusive — you cannot run both at the same time. Starting a replay with a live capture active will do nothing.

Font & Display

Font controls are on the right side of **Toolbar 2** (the BPF filter row). They apply to everything in the EtherApe window simultaneously.

CONTROL	DESCRIPTION
Size spinner	Font size 6–24. Type a value and press Enter, or click the spinbox arrows. Takes effect on the next render tick (~200ms).
Font dropdown	Choose from: monospace, Consolas, Courier New, DejaVu Sans Mono, Arial, Helvetica, Tahoma.

Font changes apply to:

- Canvas node labels and protocol legend
- Active Hosts and Active Flows treeview rows and headers
- Flow detail text panel
- All toolbar labels, buttons, and entry fields
- Status bar

Troubleshooting

Packet Capture Fails Immediately



Most common cause: not running as Administrator. Right-click the app shortcut → Run as administrator. Npcap requires elevated privileges to capture packets.

If you are running as admin and capture still fails, check:

- ▶ Npcap is installed — check Add/Remove Programs
- ▶ Try a different interface from the dropdown
- ▶ The error dialog will show the exact tshark error message

Speed Test Shows "All Attempts Failed"

- ▶ Check that `speedtest.exe` is in the application install folder
- ▶ Ensure your firewall / antivirus is not blocking speedtest.exe
- ▶ Check your internet connection
- ▶ The console output (if running with console=True) will show detailed error messages

EtherApe Shows "No Traffic"

- ▶ Make sure you clicked ► Start (not just opened the window)
- ▶ Check the selected interface has active traffic — try your Wi-Fi or Ethernet adapter
- ▶ Loopback interface only shows local traffic between processes on the same machine
- ▶ If using a BPF filter, verify the filter syntax is correct

AI Query Returns "401 Unauthorized"

- ▶ Your API key is missing or incorrect — re-enter it in the API key field
- ▶ Get a key at **console.anthropic.com** → API Keys → Create Key
- ▶ Check that you clicked `Save Key` after entering it

Report Generation Fails

- The report requires at least one entry — click + Add to Report after getting an AI response
- Check you have write permission to the chosen save location
- Try saving to the Desktop if the chosen folder fails

DNS Gauge Shows 0

The DNS gauge only updates when you click . It does not update automatically — click it whenever you want a fresh reading.

Application Freezes During Heavy Capture

The Wireshark Monitor uses a batched update system that inserts packets in groups of 150 every 100ms. If the UI feels sluggish with very high traffic, try applying a display or capture filter to reduce the packet rate.

Quick Reference

Main Dashboard

ELEMENT	ACTION
 RUN SPEED TEST NOW	Trigger an immediate speed test
 MONITOR DNS	Run a DNS resolution check now
 EXPORT DATA	Open export dialog (CSV or JSON)
 WIRESHARK	Open Wireshark Monitor window
 ETHERAPE	Open EtherApe topology window
View dial segments	Click to switch chart time window
CLR swatches (●)	Click to open colour picker for that channel

Wireshark Monitor

ELEMENT	ACTION
 Start Capture	Begin capturing on selected interface
 Stop	Stop capture
 Clear	Clear packets and delete temp PCAP
Right-click packet	Context menu: follow stream, copy IP, quick filter
 in filter	Apply display filter
Column header click	Sort by that column (click again to reverse)
 AI Query	Open AI analysis window

EtherApe Monitor

ELEMENT	ACTION
 Start /  Stop	Start or stop live packet capture
 Clear	Reset topology — clear all nodes and flows
 Open PCAP	Load a PCAP for replay
 Replay /  Stop	Start or stop PCAP replay
Click flow line	Show flow detail panel for nearest flow
Click Active Flows row	Show flow detail panel for selected row
Drag sash handles	Resize panels (canvas, hosts, flows, detail)
Size / Font controls	Adjust font for entire window (toolbar 2, right side)
Protocol dropdown	Filter displayed flows by protocol
Sort host column	Click Active Hosts column header to sort

Data File Locations

FILE	LOCATION
speedtest_data.json	%TEMP%\speedtest_monitor\ — all speed test and DNS history
speedtest_config.json	%TEMP%\speedtest_monitor\ — colour theme and settings
nm_wireshark.pcap	%TEMP%\ — current Wireshark capture (overwritten each session)
.nm_anthropic_key	%USERPROFILE%\ — saved Anthropic API key



NETWORK MONITOR USER MANUAL · VERSION 1.0

Built with Python · matplotlib · tkinter · tshark · Claude API
